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Overclocking is a process which allows you to make your processor run at a faster clock rate than what it is rated at, so that you can derive more performance out of your processor at no extra cost.

Overclocking does come with its share of risks as you are making it run at a faster speed. If done with patience and care, overclocking can boost your system's performance quite a bit, for free. Also, nowadays most of the processors come with lots of fail safes. For instance, if you have overclocked your system beyond its threshold, then it will throw up errors or hang or cause artefacts to appear in your games. These are signs that you should pay attention to and reduce the processor clock speeds.

While overclocking you need to keep four things in mind – the base clock speed,

will refrain from repeating what was already said. In fact, we thought of making this article more practical and overclocked an Intel processor. So this is a step-by-step guide to it.

Overclocking Intel Core i7-875K

Step 1

After setting the rig, we enter the BIOS by pressing Del. As we are using the MSI P55 motherboard, we will have to enter the Cell Menu to tweak the processor settings. For ASUS Motherboards you will have to enter Ai Tweaker and so on.

We disable the Turbo Boost, as we do not want the processor to overclock on its own, as it will not give us a clear idea of the performance jump we can expect. Similarly SpeedStep (EIST) was also disabled as it is not really needed for overclocking. The Base Clock (BCLK) in our case was 133 MHz and Multiplier was 22x. The RAM

timings were set to 7-7-7-20 for the 2x2GB Corsair XMS3 DDR3 memory.

Step 2

As we were just doing basic overclocking, using only stock components, we did not touch the voltages. We started with increasing the multiplier count from 22x to 23x, which pushed the processor frequency to 3.06 GHz. We saved these settings by pressing F10. On rebooting the system, we checked all the temperatures in the idle state as well as on load. We ran a few benchmarks. At no point did we notice any crashing or blue screen. The system was certified stable at 3.06 GHz.



Overclocked status

Step 3

We repeated Step 2 and this time increased the multiplier to 24x. This process continued, till we reached a point where increasing the multiplier beyond 26x resulted in an unstable system. So on the next reboot, we reduced the multiplier to the last stable setting and started playing with the base clock frequency. For the purpose of this article, we stretched the core clock to 3.53 GHz and ran all the benchmarks at this frequency. The table below will give you an idea of the readings

Step 4

Be sure to keep a tab on the temperatures. It should not overheat at any point of time, as the transistors working at faster clock speeds will generate more heat than at stock speeds.

Finally, just mere overclocking is not enough. The system should run comfortably at the overclocked settings. Only then can it be called a successful overclock.

As you see we got quite a performance boost not just with games, but with productivity tasks like rendering, encoding, etc. So what are you waiting for?

Remember, you can even overclock a multiplier locked processor by tweaking the base clock frequency. But remember that running a stable system is more important than extreme overclocking.

Overclocking your processor

Stretching your system performance for free

the multiplier, the latency timings and the temperature. Base clock speed is the speed at which the processor runs and when multiplied with the multiplier, will give you the core clock speed of the processor.

To read a detailed article on the process, we would like you all to read the Overclocking Guide we had published in November 2010 (also found here <http://bit.ly/b0XYMw>). We

Processor: Intel Core i7 875K @ 2.93 GHz
Memory: Corsair XMS 2x2GB DDR3 @ 1666MHz
Motherboard: MSI P55A GD65
Graphic Card: Palit GTX 460
Power Supply: Corsair HX620W
Monitor: Viewsonic VX2250mw
Hard Disk: WD Velociraptor 300GB

Benchmarks	Stock Speed (2.93 GHz)	Overclocked (3.53 GHz)
Cinebench R10 (multiple CPU)	11615	13718
Unigine Heaven 2.0	35 fps	37.8 fps
DixX Encoding (100 MB VOB to DivX conversion)	32 seconds	23 seconds
Doom 3	310 fps	367.4 fps
Resident Evil 5	99 fps	103 fps